

Yuquan Chen

412-475-6091 | altheac297@gmail.com | linkedin.com/in/yuquan-althea-chen

EDUCATION

Carnegie Mellon University

Master of Information Systems Management (MISM)

August 2025 – December 2026
Pittsburgh, PA

Relevant Courses: Data-Focused Python (A), Managing Disruptive Technologies (A–), NLP (B+), Stats (A–)

Nanjing University

Bachelor of Information Management & Information Systems

September 2021 – June 2025
Nanjing, China

GPA: 4.32/5.0 | Relevant Courses: Practical ML & Deep Learning (94), Social Network Analysis (95.2), Text Data Management (93), Big Data Systems (92), Operations Research (89)

WORK EXPERIENCE

Whelix — AI Robotics Startup (Biotech)

AI Engineer Intern

May 2026 – Present
Boston, MA

- Designed and shipped a production multi-agent AI platform for a robotics startup from the ground up, built directly on the raw **Gemini SDK** (no agent framework) to retain full control over the tool-use loop — architecting a single orchestrator agent that dynamically delegates to **three** specialized sub-agents at runtime via a dedicated routing tool, re-verifying authorization on every delegation to gate two CEO-only specialists entrusted with live sales-pipeline and fundraising data, deployed on **Google Cloud Run**.
- Integrated the platform across **three** external systems (Notion, Confluence, and Google Workspace — Gmail, Calendar, and Chat) via **26 custom tool functions** for structured data read/write, and engineered a two-layer memory architecture — live structured-fact writes plus GCS-backed conversation logging with nightly LLM-driven reconciliation — enabling stateless webhook calls to sustain coherent multi-turn context.

BASF Polyurethane Specialties (China) Co., Ltd.

Digitalization Intern, Asia-Pacific Monomers

May 2024 – February 2025
Shanghai, China

- Identified a critical intelligence gap in BASF's MDI/TDI pricing process: marketing analysts were manually monitoring 3–4 news sources across two languages to track demand signals — an error-prone process feeding VP-level pricing decisions in a market where BASF holds significant pricing authority in China.
- Designed and deployed an **end-to-end LLM-powered (GPT-4o) market intelligence engine** that automated bilingual news monitoring, extraction, and synthesis — replacing the manual workflow entirely and delivering structured weekly and monthly insight reports directly consumed by pricing decision-makers.
- Partnered with marketing stakeholders to ensure output format matched executive reporting requirements, driving adoption from day one and eliminating the need for manual news triage across the team.

SELECTED PROJECTS & AWARDS

Thesis: Decision-Focused Uplift Modeling for AUUC Maximization

February – May 2025

- Designed and implemented Decision-Focused Uplift Modeling (DFUM), a training framework that closes the gap between prediction accuracy and AUUC-measured ranking decision quality in individual treatment effect estimation — combining a binary cross-entropy prediction loss with a differentiable, softmax-weighted decision loss that directly optimizes the downstream treatment-ranking objective, motivated by an integer linear programming formulation of budget-constrained treatment assignment
- Benchmarked DFUM against S-Learner, X-Learner, UpliftRank, and Generalized Random Forests across a 13M-row public ad-incrementality dataset (Criteo), achieving an average 2%+ AUUC improvement over 20 independent seeds per configuration while maintaining stable performance across a swept trade-off hyperparameter

CCF Big Data & Computing Intelligence Contest (1,557 participants)

October – December 2023

- Led a team to a 2nd-place finish (out of 1,557 teams) in the CCF Big Data & Computing Intelligence Contest (BDCI 2023), designing an end-to-end forecast-stock-allocate decision pipeline for an omnichannel fresh-grocery inventory optimization challenge
- Built a demand-forecasting model on top of the Kaggle M5 competition's top-1 approach — engineering price, promotion, sales-lag, and calendar features, then training a multi-window ensemble (30/90/365-day and full-history LightGBM models) to jointly capture short- and long-term sales patterns across SKUs with heterogeneous listing dates
- Designed a nonlinear safety-stock formula that allocates proportionally more buffer inventory to low-volume, harder-to-forecast SKUs, and a greedy heuristic for front-/back-store inventory allocation under service-level constraints — trading exact profit-optimal solutions (computationally intractable within the contest's time limit) for fast, near-optimal heuristics

TECHNICAL SKILLS

Analytics & AI: LLM Agent Systems (LangGraph, AutoGen), Machine Learning, Uplift Modeling, Causal Inference, NLP, Time Series Forecasting

Tools & Visualization: Python (Pandas, NumPy, Scikit-learn), SQL, Power BI, KNIME, R

Languages: English (Professional Working), Mandarin Chinese (Native), Spanish (Elementary)